

Week 1 Module – Programming Practice (3rd Year)

Date: 10/06/2026

Language Used

Students can use either C++ or Java Programming Language to complete all tasks.

Topics to be Covered:

1. Star Patterns

1. Square 2. Hollow Square 3. Triangle 4. Inverted Triangle

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5. Pyramid 6. Diamond 7. Half Diamond 8. Half Diamond Inverted

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2. Number Patterns

1. Square 2. Incrementing 3. Right Triangle 4. Inverted Diamond

1 1 1 1 1	1 2 3 4 5	1	3 3 3 3 3
1 1 1 1 1	1 2 3 4 5	1 2	4 4 4 4
1 1 1 1 1	1 2 3 4 5	1 2 3	5 5 5
1 1 1 1 1	1 2 3 4 5	1 2 3 4	6 6
1 1 1 1 1	1 2 3 4 5	1 2 3 4 5	7

5. Sandwich Pattern

```
3 3 3 3 3
4 4 4
5
4 4 4
3 3 3 3 3
```

6. Incrementing Diamond Pattern

```
3
4 4
5 5 5
6 6 6 6
7 7 7 7 7
6 6 6 6
5 5 5
4 4
3
```

Task Requirements

Students are required to:

- Write the **logic and code** for all given patterns.
- Use **either C++ or Java** for implementation.
- Execute every program and verify the output.

PDF Submission Requirements

Students must prepare a **single PDF document** containing:

1. Handwritten Work

- Handwritten logic/approach for each pattern
- Handwritten code in notebook

2. Executable Code Proof

- Screenshot of successfully executed code/output for every program

GitHub Submission

- Push the final PDF to GitHub.
- **Submission Date: 14/06/2026**
- Repository name format: **3rd Yr Week 1 Module**

Important Instructions

- ✓ Handwritten notes and code are mandatory
- ✓ Output screenshots must be clearly visible
- ✓ Maintain proper sequence and neatness in the PDF
- ✓ Incomplete submissions will not be accepted